MILITARY SPECIFICATION

RECHARGING UNIT, CARBON DIOXIDE, PUMPING METHOD

1. SCOPE

1.1 This specification covers a high pressure liquid carbon dioxide recharger unit for filling standard high pressure fifty (50) pound carbon dioxide cylinders (storage).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CC-M-641	Motor, Alternating Current
QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-E-489	Enamel; Gloss, Synthetic (For Exterior and Interior Surfaces)
TT-P-636	Primer Coating, Synthetic, Wood and Ferrous Metal
TT-V-51	Varnish, Asphalt
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner

Military

MIL-P-116	Preservation, Methods of
MIL-C-490	Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings
MIL-S-4511	Storage Unit, Low Pressure, Carbon Dioxide, Type A-1
MIL-S-5002	Surface Treatments (Except Priming and Painting) for Metal and Metal Parts in Aircraft
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-N-7513	Nomenclature Assignment and Nameplate Approval, Contractors Method for Obtaining
MIL-E-7729	Enamel, Gloss, for Aircraft Use

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MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports, Preparation of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-143	Order of Precedence for the Selection of Specifications and Standards
MS33586	Metals - Definitions of Dissimilar
MS35612	Primer, Synthetic, Lacquer Resisting
MS35613	Primer, Zinc Chromate, for Aircraft

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. The preproduction model is defined as a production model of the product which the manufacturer shall produce in fulfillment of the contract or order and prior to submission of any complete units for final acceptance, the contractor shall furnish a complete assembly to determine conformance to the requirements contained herein. Examination and tests shall be as specified herein. Approval of the preproduction model by the activity concerned shall not relieve the contractor of his obligation to supply equipment in accordance with the requirements contained herein. Any changes or deviations of the production units from the preproduction model shall be subject to the approval of the contracting officer.

3.2 Components. The recharging unit shall consist of the following major components: a compressor; drive motor; control panel; and housing.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified

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herein shall be selected in accordance with MIL-STD-143, except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts like screws, bolts, nuts, cotter pins, etc, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts lists and, if practicable, on the contractor's drawings.

3.3.2 Standard parts. With the exception in 3.3.1 AN and MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials.

3.4.1 Dissimilar metals. Dissimilar metals shall be as defined in Standard MS33586. Unless suitably protected against electrolytic corrosion by means of protective coating, intimate contact of dissimilar metals shall be avoided. When in contact with dissimilar metals, aluminum parts shall be anodized in accordance with Specification MIL-A-8625, and steel parts shall be cadmium plated in accordance with Specification QQ-P-416, type II or III, as applicable, and of a suitable class.

3.4.2 Protective treatment. When materials are used in construction of the recharging unit that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design. The recharging unit shall be designed so that it is capable of filling two cylinders simultaneously at a minimum total flow rate of 25 pounds per minute of liquid carbon dioxide from a 300 psi source to between 600 and 1000 psi inclusive operating discharge pressure. The recharging unit is primarily intended for use in conjunction with, but not an integral part of, the storage unit, low pressure carbon dioxide type A-1 (MIL-S-4511) and must be entirely suitable for this application.

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3.6 Construction. The recharging unit and all components thereof shall be capable of withstanding the strains, jars, twists, vibrations, and other conditions incident to shipping, storage, installation and service.

3.7 Performance. The recharging unit shall be capable of starting and operating at rated capacity at -65 degrees F and +130 degrees F ambient temperatures when the equipment temperature prior to starting is at the former and latter temperatures respectively.

3.7.1 Housing. The recharging unit shall be fully contained in one unit, and shall be enclosed in a steel housing of welded construction and braced with sufficient members to provide adequate rigidity when plated upon a flat, level, rigid base. Lifting lugs shall be provided at the base to carry the weight of the complete recharging unit.

3.7.2 Compressor. A compressor of adequate capacity to maintain characteristics as specified in 3.5 shall be installed. The compressor unit shall be air-cooled.

3.7.2.1 Compressor drive motor. The drive motor shall be a 5 hp, 1720 rpm, 3 phase, 220/440 volt, 60 cycle induction type in accordance with CC-M-641. The electric motor shall be provided with thermal overload protection.

3.7.2.2 The motor control switch shall be manual, push-button stop-start across the line switch equipped with thermal motor overload protection elements.

3.7.3 Safety devices. The safety devices shall consist of a frangible disc assembly and spring-loaded pop valves. All safety devices shall be located and arranged to afford maximum protection to the operator in the event of device actuation.

3.7.3.1 Frangible disc assembly. Frangible disc assembly shall be provided between the compressor and the regulating valve to protect the compressor from possible damage due to excessive pressure.

3.7.3.2 Spring-loaded pop valves. The spring-loaded pop valves shall be provided with seats to insure tightness.

3.7.3.3 Inlet line pop valve. The spring-loaded pop valve on the inlet line shall relieve at 400 psi.

3.7.3.4 Discharge line pop valve. The spring-loaded pop valve on the discharge line shall be designed similarly to the valve on the inlet lines for protection against excessive pressure.

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3.7.4 Control panel. The control panel shall be designed so that complete operation may be readily accomplished by one operator. The control panel shall be equipped with all manual controls such as unit starting and stopping facility, shut-off valves, and pressure regulating valves necessary for filling pressure adjustment.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of Specification MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.9 Dimensions.

3.9.1 The maximum dimensions of the recharging units shall not exceed the following:

a. Length - 43 inches

b. Width - 23 inches

c. Height - 40 inches

3.10 Weight. The total weight of the recharging unit shall not exceed 620 pounds. This weight excludes cylinders, lines to and from storage unit, connecting hoses and weighing scales.

3.11 Finishes and protective coatings.

3.11.1 General. All metal parts shall be cleaned in accordance with Specification MIL-S-5002. After treatment and application of primer specified in 3.11.2 and 3.11.3, the following shall be accomplished. The unit shall be finished with at least two coats of enamel conforming to Specification MIL-E-7729. Base parts shall be finished with at least two coats of varnish conforming to Specification TT-V-51. Colors shall be in accordance with FED-STD-595.

3.11.2 Ferrous parts. Sheet and other ferrous metal parts shall be treated in accordance with Specification MIL-C-490, grade 1, prior to being primed. The primer shall conform to Specification TT-P-636 or MS35612.

3.11.3 Aluminum parts. Aluminum parts shall be given a corrosion preventive chemical film conforming to Specification MIL-C-5541. Following the corrosion preventative treatment, two coats of primer conforming to Specification MS35613 shall be applied.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

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3.12.1 Additional marking. Markings shall be glossy white enamel conforming to Specification TT-E-489.

3.12.1.1 Center of gravity. The center of gravity shall be permanently and legibly marked on both the length-height face and the width-height face of the recharging unit.

3.12.1.2 Instruction plates. All necessary instructions and cautions shall be stencilled in letters of applicable size for ease of reading in the vicinity of the controls or mechanisms requiring special attention.

3.13 Nomenclature. Establishing of nomenclature and type designation shall be in accordance with Specification MIL-N-7513, subject to the approval of the Air Force.

3.14 Workmanship

3.14.1 General. The recharging unit, including all parts and accessories shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts, et cetera.

3.14.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions or tolerances may affect the interchangeability, operation or performance of the recharging unit, they shall be held or limited accordingly.

3.14.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" shall mean that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.14.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall develop adequate strength in the parts connected.

3.14.5 Cleaning. The recharging unit shall be thoroughly cleaned and loose, spattered or excess solder, metal chips and other foreign material removed during and after final assembly.

3.15 Such technical publications as specified shall be furnished (see 6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Classification of tests. The inspection and testing of recharging units shall be classified as follows:

a. Acceptance tests ..... See 4.3, 4.4, 4.5

b. Preproduction testing ..... See 4.6

4.3 Acceptance tests.

4.3.1 Individual tests. Each recharging unit shall be subjected to the following tests.

4.3.1.1 Examination of product. The recharging unit shall be subjected to a visual inspection to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions and marking.

4.4 Test conditions.

4.4.1 Atmosphere. The performance ability of the recharging unit when affected by atmospheric conditions and unless otherwise specified, shall be demonstrated at, or corrected for, standard sea level atmospheric conditions of 29.92 inches of mercury barometric pressure and ambient temperature of 60 degrees F.

4.4.2 Apparatus. Apparatus used in conjunction with testing specified herein shall be, insofar as practicable, of laboratory precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy.

4.4.3 All items submitted for test shall be properly prepared by the contractor for operation; all items of equipment shall be in correct adjustment, lubricated, and serviced for immediate operation.

4.5 Test methods. Test methods shall be so conducted that requirements of 4.3, 4.4, 4.5 and 4.6 will be met.

4.6 Preproduction tests. Preproduction tests shall be performed contractor in accordance with the following paragraphs.

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4.6.1 Preproduction test samples. One recharging unit, when specified in the contract, shall be submitted to the procuring activity, at the time the preproductinn sample test report (see 6.2.3) is submitted for approval. The samples will be used by the procuring activity for:

a. A review of the mechanical construction of the recharging unit.

b. Performance of any tests which may be deemed necessary after a review of the contractor's test report.

c. Government reference purposes.

4.6.2 The preproduction test shall consist of the acceptance tests and the following tests.

4.6.2.1 Features and components. The recharging unit shall be critically inspected to determine that all required components and features are included and properly installed.

4.6.2.2 Functional test. The input side of the recharging unit shall be connected to a source of liquid carbon dioxide under a minimum pressure of 300 psi. The output of the recharger shall be connected to two fifty pound empty cylinders. The recharger shall fill the two cylinders simultaneously at a total flow rate of at least 25 pounds per minute. Verify that a charging pressure of 600, 800 and 1,000 psi can be maintained.

4.6.2.3 Low temperature test. The recharging unit shall be placed in an ambient air temperature of minus 65 degrees F for a period of not less than 24 hours. At the end of the 24 hour period, while maintaining this temperature, the unit shall be subjected to the test specified in 4.6.2.2.

4.6.2.4 High temperature test. The recharging unit shall be placed in an ambient air temperature of plus 130 degrees F for a period of not less than 24 hours. At the end of the 24 hour period, while maintaining this temperature, the unit shall be subjected to the test specified in 4.6.2.2.

4.6.2.5 Mechanical checks. A critical inspection shall be made of all components to determine their operational ability and any damage or undue wear incurred at the end of the preproduction testing program. Teardown and parts measurement shall be made only in those cases where service life is in question.

4.6.3 Safety devices. The safety devices shall be tested to conform to the requirements of 3.7.3.

4.6.4 Determination of the center of gravity. The center of gravity of the recharging unit with all equipment installed shall be determined. The recharging unit shall be tilted at

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least 10 degrees fore and aft during weighing operations and the intersection of the respective vertical lines obtained.

4.6.5 Electrical equipment. All electrical equipment shall be operated to insure satisfactory operation.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Recharging units shall be processed in accordance with MIL-P-116, Method IIA. The compressor crankcase shall be drained. A tag shall be affixed to the compressor indicating that the crankcase is drained and must be filled with the manufacturer's recommended lubricant prior to operation.

5.1.2 Level C. Recharging units shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 Level A. Recharging units shall be packed in skidded export type boxes conforming to PPP-B-601, or PPP-B-621. Closure and strapping shall be in accordance with the applicable container specification or appendix.

5.2.2 Level B. Recharging units shall be packed in skidded domestic type boxes conforming to PPP-B-601 or PPP-B-621. Closure and strapping shall be in accordance with the applicable container specification or appendix.

5.2.3 Level C. Recharging units shall be packed in a manner to assure safe delivery at the lowest cost for packing and freight charges combined.

5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Use. The recharging unit is primarily intended for use in conjunction with, but not as an integral part of, the storage unit, low pressure carbon dioxide, of Specification MIL-S-4511, type A1, and must be entirely suitable for this application.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

c. Technical publications required (see 3.15).

6.3 Preproduction sample. One charging unit will be required as a preproduction sample to be subjected to preproduction tests by the contractor to determine compliance with requirements of this specification.

6.4 Preproduction test report. After completion of the preproduction tests by the contractor, three copies of a complete report shall be furnished the procuring activity. The preproduction test report shall be in accordance with Specification MIL-T-9107.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permissions to manufacture, use, or sell any patented invention that may in any way be related thereto.

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